

13.2 How Derivatives Affect the Shape of a Graph

Definition:

$f(x)$ is increasing on the interval I if $f(x_1) < f(x_2)$ for $x_1 < x_2 \in I$.

Increasing/Decreasing Test

- (a) If $f'(x) > 0$ on an interval, then f is increasing on the interval.
- (b) If $f'(x) < 0$ on an interval, then f is decreasing on the interval.

The First Derivative Test

Suppose that c is a critical number of a continuous function f .

- (a) If f' changes from positive to negative at c , then f has a local maximum at c .
- (b) If f' changes from negative to positive at c , then f has a local minimum at c .
- (c) If f' does not change sign at c (for example, if f' is positive on both sides of c or negative on both sides), then f has no local maximum or minimum at c .

Example 2.1

Prove the Increasing/Decreasing Test.

Pf: Suppose that $f'(x) > 0$ on I .

For $x_1 < x_2$, $x_1, x_2 \in I$, $f(x_2) - f(x_1) \stackrel{\text{M.V.T.}}{=} \underbrace{f'(c)}_{+} \underbrace{(x_2 - x_1)}_{+} > 0$ for some $c \in (x_1, x_2) \subset I$
 $\Rightarrow f(x_2) > f(x_1) \Rightarrow f(x)$ is increasing on I .

Example 2.2

$f(x) = 2x^3 + 3x^2 - 12x + 7$. Find intervals of increase/decrease and local extreme values of $f(x)$.

Sol: $f'(x) = 6x^2 + 6x - 12 = 6(x^2 + x - 2) = 6(x+2)(x-1)$

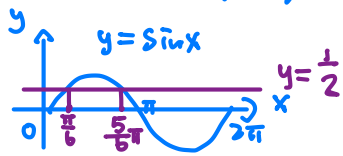
		local max?		local min?		
		-2		1		
$x+2$	-		+		+	
$x-1$	-		-		+	
f'	+		-		+	

$\Rightarrow f'(x) > 0$ on $(-\infty, -2) \cup (1, \infty)$
 $f'(x) < 0$ on $(-2, 1)$.
 $\Rightarrow f$ is increasing on $(-\infty, -2), (1, \infty)$
 f is decreasing on $(-2, 1)$.

Example 2.3

$f(x) = x + 2 \cos x$, $0 \leq x \leq 2\pi$ Find intervals of increase/decrease and local extreme values of $f(x)$.

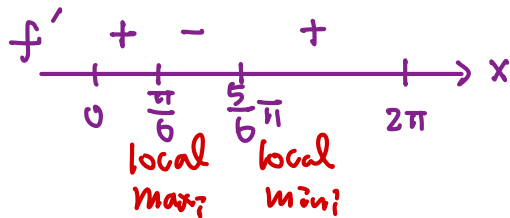
Sol: $f'(x) = 1 - 2 \sin x$.



$$f'(x) > 0 \text{ on } (0, \frac{\pi}{6}) \cup (\frac{5\pi}{6}, 2\pi)$$

$$f'(x) < 0 \text{ on } (\frac{\pi}{6}, \frac{5\pi}{6})$$

$\Rightarrow f$ is increasing on $(0, \frac{\pi}{6}) \cup (\frac{5\pi}{6}, 2\pi)$
 „ decreasing „ $(\frac{\pi}{6}, \frac{5\pi}{6})$.

**Example 2.4**

$f(x) = x^{\frac{1}{3}}(x-6)^{\frac{2}{3}}$. Find intervals of increase/decrease and local extreme values of $f(x)$.

Sol: $f'(x) = \frac{1}{3} x^{-\frac{2}{3}}(x-6)^{\frac{2}{3}} + x^{\frac{1}{3}} \cdot \frac{2}{3} (x-6)^{-\frac{1}{3}} = \frac{(x-6)^{\frac{2}{3}}}{3x^{\frac{2}{3}}} + \frac{x^{\frac{1}{3}} \cdot 2}{3(x-6)^{\frac{1}{3}}}$

$$= \frac{(x-6) + 2x}{3x^{\frac{2}{3}}(x-6)^{\frac{1}{3}}} = \frac{(x-2)}{x^{\frac{2}{3}}(x-6)^{\frac{1}{3}}}$$



$$f'(x) > 0 \text{ on } (-\infty, 0) \cup (0, 2) \cup (6, \infty)$$

$$f'(x) < 0 \text{ on } (2, 6)$$

Critical number: $x = 0, 2, 6$

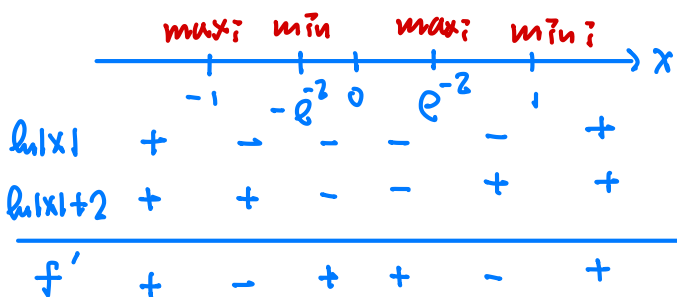
↑ local max?
 ↑ local min?

Example 2.5

$f(x) = x(\ln|x|)^2$. Find intervals of increase/decrease and local extreme values of $f(x)$.

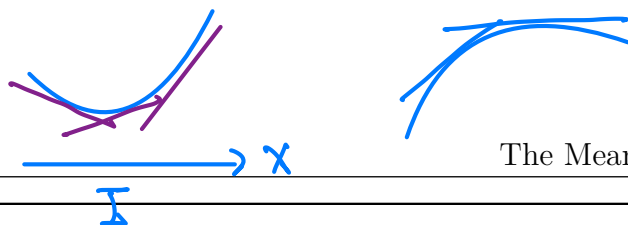
Sol: $f'(x) = (\ln|x|)^2 + x \cdot 2 \cdot \ln|x| \cdot \frac{1}{x} = (\ln|x|)^2 + 2 \ln|x|$

$$= \ln|x| (\ln|x| + 2)$$



$$f'(x) > 0 \text{ on } (-\infty, -1) \cup (-e^{-2}, 0) \cup (0, e^{-2}) \cup (1, \infty)$$

$$f'(x) < 0 \text{ on } (-1, -e^{-2}) \cup (e^{-2}, 1)$$

**Definition:**

If the graph of f lies above all of its tangents on an interval I , then it is called **concave upward** on I . If the graph of f lies below all of its tangents on I , it is called **concave downward** on I .

Theorem:

$f(x)$ is concave upward on an open interval I if and only if $f'(x_1) \leq f'(x_2)$ for all $x_1 < x_2$, $x_1, x_2 \in I$.

Concavity Test

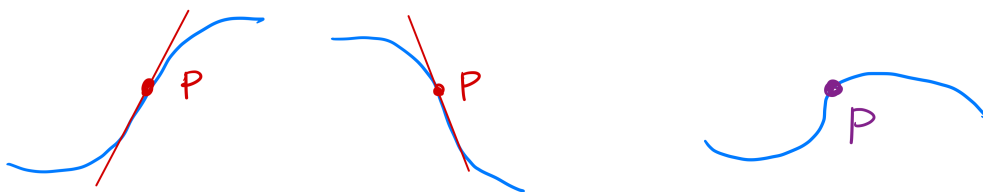
- (a) If $f''(x) > 0$ for all x in I , then the graph of f is concave upward on I .
- (b) If $f''(x) < 0$ for all x in I , then the graph of f is concave downward on I .

Definition:

A point P on a curve $y = f(x)$ is called an **inflection point** if f is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at P .

Remark:

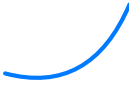
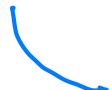
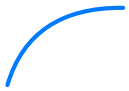
From the definition, we have assumed that $f(x)$ is continuous at an inflection point and is differentiable on its both sides. But $f(x)$ may not be differentiable at an inflection point.

**Property:**

If $P = (a, f(a))$ is an inflection point of $y = f(x)$ and $f(x)$ is differentiable at $x = a$, then

1. $f'(x)$ has local extreme value at $x = a$.
2. The tangent line of $y = f(x)$ at P crosses the graph $y = f(x)$ there.
3. If $f''(a)$ exists, then $f''(a) = 0$.

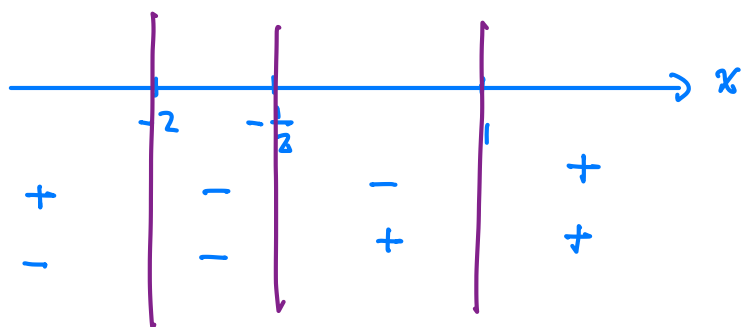
Sketch the graph of a Function

	$f' > 0$ (Increasing)	$f' < 0$ (Decreasing)
$f'' > 0$ (Concave upward)		
$f'' < 0$ (Concave downward)		

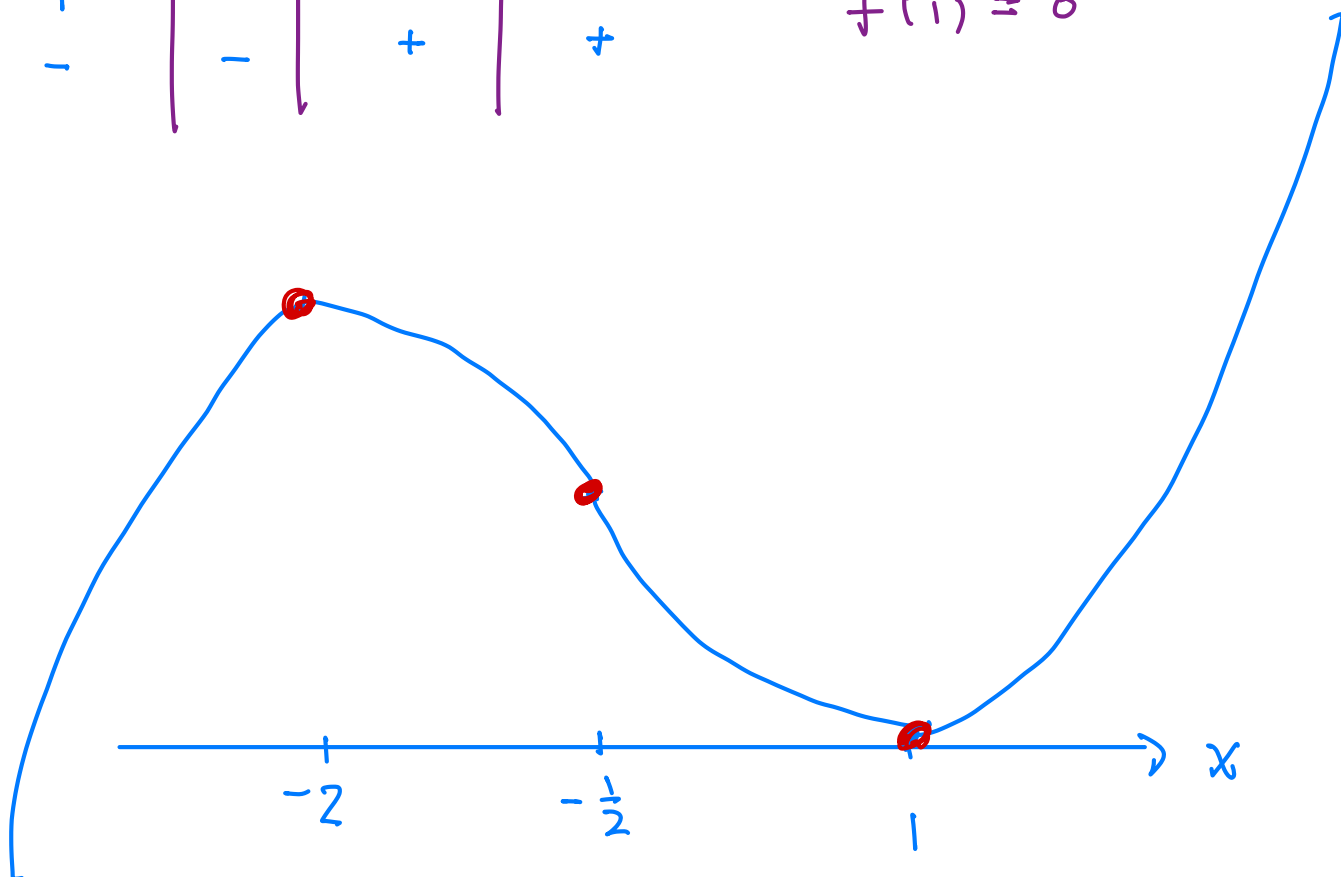
Example 2.6

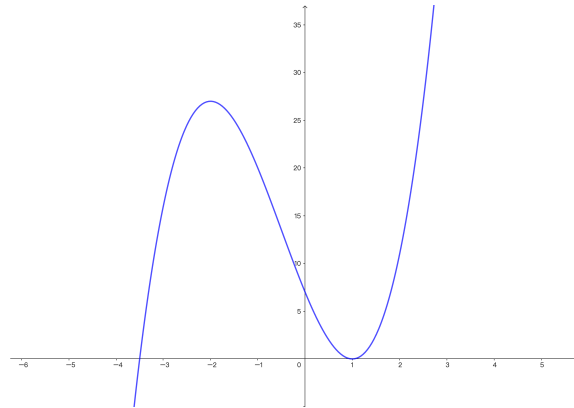
Determine Concavity of $f(x) = 2x^3 + 3x^2 - 12x + 7$ and sketch the graph of $f(x)$.

sol: $f'(x) = 6x^2 + 6x - 12$ $f''(x) = 12x + 6$, $f'' > 0$ on $(-\frac{1}{2}, \infty)$
 $y = f(x)$ is concave upward on $(-\frac{1}{2}, \infty)$ $f'' < 0$ on $(-\infty, -\frac{1}{2})$.
 ∴ downward on $(-\infty, -\frac{1}{2})$.



$$\begin{aligned} f(-2) &= 27 \\ f(-\frac{1}{2}) &= \frac{27}{2} \\ f(1) &= 0 \end{aligned}$$





Determine Concavity of $f(x) = x^{\frac{1}{3}}(x - 6)^{\frac{2}{3}}$ and sketch the graph of $f(x)$.

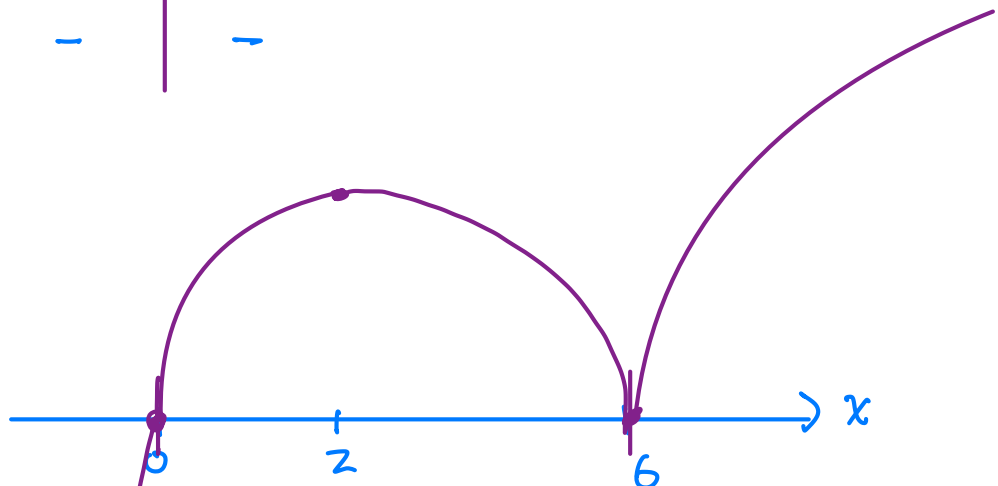
$$\begin{aligned} \text{sol: } f'(x) &= \frac{x^{-2}}{x^{2/3} (x-6)^{1/3}} = (x-2) x^{-5/3} (x-6)^{-1/3} \\ f''(x) &= x^{-2/3} (x-6)^{-1/3} - \frac{2}{3} (x-2) x^{-5/3} (x-6)^{-1/3} - \frac{1}{3} (x-2) x^{-2/3} (x-6)^{-4/3} \\ &= \frac{1}{x^{2/3} (x-6)^{1/3}} - \frac{2(x-2)}{3 x^{5/3} (x-6)^{1/3}} - \frac{x-2}{3 x^{2/3} (x-6)^{4/3}} \\ &= \frac{3x(x-6) - 2(x-2)(x-6) - (x-2)x}{3 x^{5/3} (x-6)^{4/3}} = \frac{-8}{x^{5/3} (x-6)^{4/3}} \end{aligned}$$

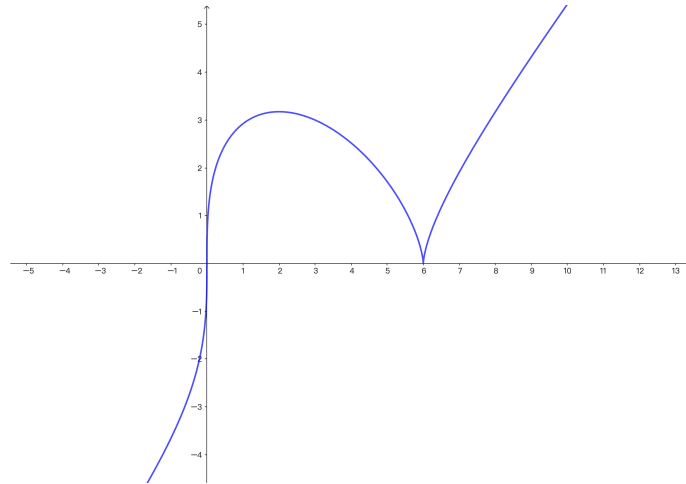
$$f'(x) > 0 \text{ on } (-\infty, 0) \quad f'(x) < 0 \text{ on } (0, \infty).$$

	$x < 0$	$0 < x < 2$	$2 < x < 6$	$x > 6$
f'	+	+	-	+
f''	+	-	-	-

$$f(0) = 0 = f(6)$$

$$f(x) = 2^{5/3}$$



**Example 2.8**

Sketch the curve $y = e^{\frac{1}{x}}$ Let $f(x) = e^{\frac{1}{x}}$.

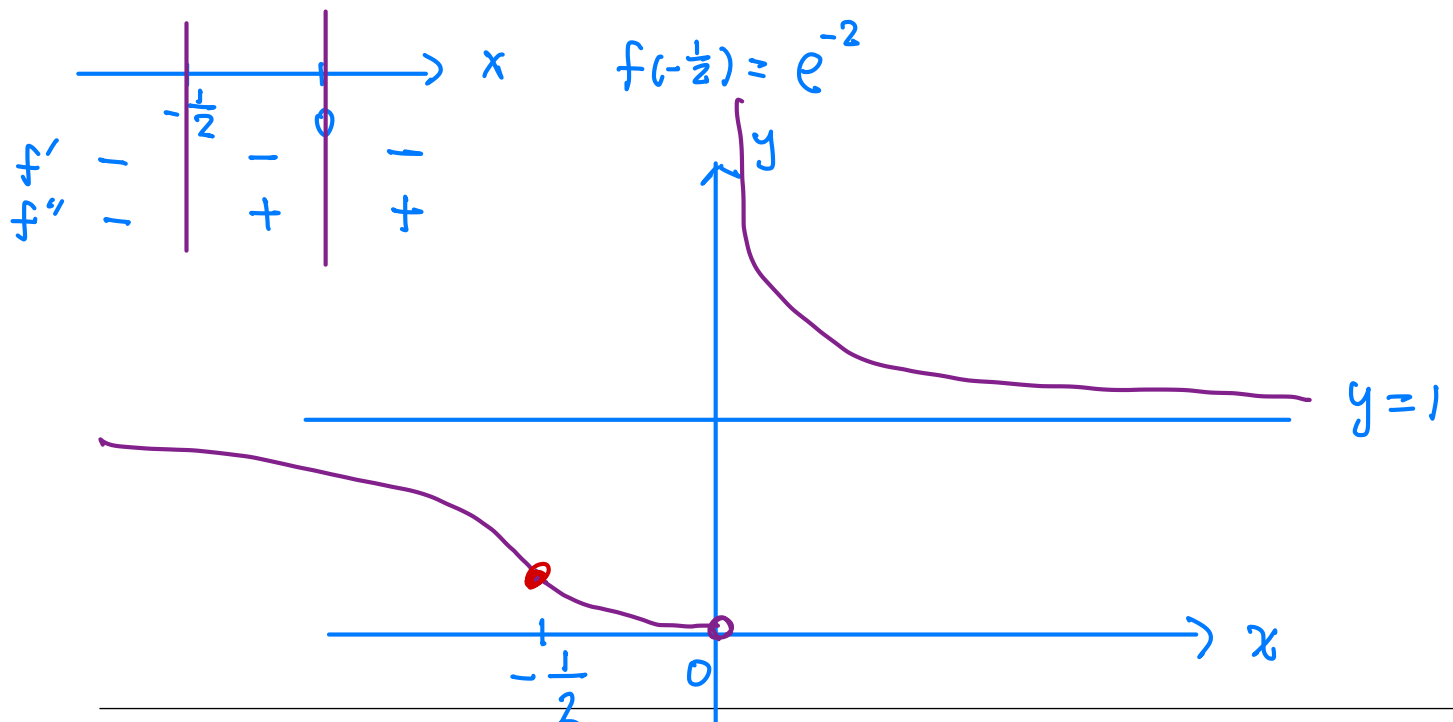
Sol: $\text{Dom}(f) = \mathbb{R} \setminus \{0\}$ $\lim_{x \rightarrow 0^+} f(x) = \infty$ $\Rightarrow x=0$ is a v.A.

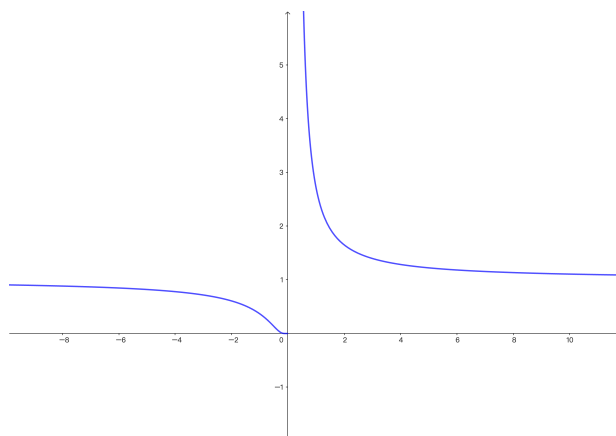
$\lim_{x \rightarrow 0^-} f(x) = 0$. $\lim_{x \rightarrow \pm\infty} e^{\frac{1}{x}} = e^0 = 1$ $\Rightarrow y=1$ is a H.A.

$$f'(x) = e^{\frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right) < 0 \text{ on } (-\infty, 0) \cup (0, \infty)$$

$$f''(x) = \underline{e^{\frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right) \cdot \left(-\frac{1}{x^2}\right)} + \underline{e^{\frac{1}{x}} \cdot \frac{2}{x^3}} = e^{\frac{1}{x}} \left(\frac{1+2x}{x^4}\right)$$

$$f''(x) < 0 \text{ on } (-\infty, -\frac{1}{2}), \quad f''(x) > 0 \text{ on } (-\frac{1}{2}, 0) \cup (0, \infty)$$



**Example 2.9**

Let $f(x) = 3 \cos^2 x + \sin x$, $x \in [-\pi, \pi]$.

- Find the intervals of increase or decrease.
- Find the intervals of concavity.
- Sketch the graph of $f(x)$.

The Second Derivative Test

Suppose f'' is continuous near c .

- If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
- If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .

